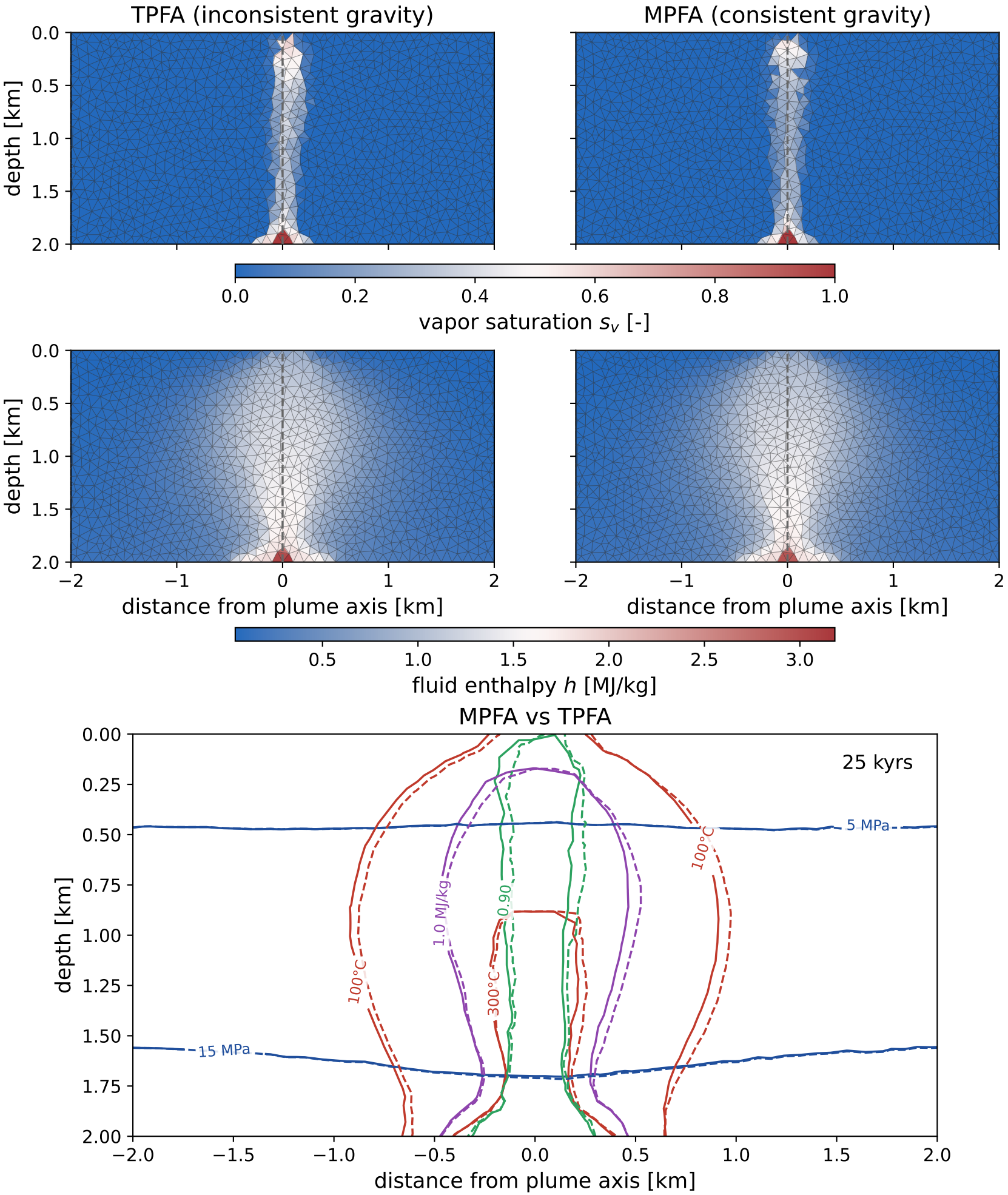

TPFA (inconsistent gravity)
MPFA (consistent gravity)
depth [km]
0.0
0.5
1.0
1.5
2.0
0.0
0.2
0.4
0.6
0.8
1.0
vapor saturation s_v [-]
−2
−1
0
1
2
distance from plume axis [km]
0.5
1.0
1.5
2.0
2.5
3.0
fluid enthalpy h [MJ/kg]
MPFA vs TPFA
25 kyrs
5 MPa
15 MPa
100°C
1.0 MJ/kg
0.90
300°C
100°C
−2.0
−1.5
−1.0
−0.5
0.0
0.5
1.0
1.5
2.0
0.00
0.25
0.50
0.75
1.00
1.25
1.50
1.75
2.00
Pressure
Temperature
Liquid saturation
Enthalpy
MPFA
TPFA